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Numerical Methods

Lab 7

Introduction

In this lab, I studied numerical integration methods. The main aim was to approximate definite integrals by using different numerical methods and compare the results with the exact analytical value.

The methods used in this lab are:

- left rectangle method,
- right rectangle method,
- midpoint rectangle method,
- trapezoidal method.

First, I calculated the integral of x^2 from 1 to 4 manually. Then I checked the same calculations by using the given Python code in lab7.ipynb. After that, I used the same numerical methods to calculate the integral of e^{-x^2} from -1 to 3 and compared how the accuracy changes when the number of intervals is increased.

Exercise 1 - Manual Calculation of the Integral

The first integral is:

$$\int_1^4 x^2 dx$$

The function is:

$$f(x) = x^2$$

The given values are:

$$\begin{aligned} x &= 1, 2, 3, 4 \\ y &= 1, 4, 9, 16 \end{aligned}$$

There are three intervals:

$$[1, 2], [2, 3], [3, 4]$$

The width of each interval is:

$$h = 1$$

Analytical Solution

First, I calculated the exact value of the integral analytically.

$$\begin{aligned}\int_1^4 x^2 dx &= \left[\frac{x^3}{3} \right]_1^4 \\&= \frac{4^3}{3} - \frac{1^3}{3} \\&= \frac{64}{3} - \frac{1}{3} \\&= \frac{63}{3} \\&= 21\end{aligned}$$

So, the exact value is:

$$I = 21$$

Left Rectangle Method

For the left rectangle method, the function values from the left side of each interval are used.

The left points are:

$$x = 1, 2, 3$$

The function values are:

$$f(1) = 1, f(2) = 4, f(3) = 9$$

So, the approximation is:

$$\begin{aligned}I_L &= h[f(1) + f(2) + f(3)] \\I_L &= 1(1 + 4 + 9) \\I_L &= 14\end{aligned}$$

The error is:

$$Error = |21 - 14| = 7$$

Right Rectangle Method

For the right rectangle method, the function values from the right side of each interval are used.

The right points are:

$$x = 2, 3, 4$$

The function values are:

$$f(2) = 4, f(3) = 9, f(4) = 16$$

So, the approximation is:

$$\begin{aligned} I_R &= h[f(2) + f(3) + f(4)] \\ I_R &= 1(4 + 9 + 16) \\ I_R &= 29 \end{aligned}$$

The error is:

$$Error = |21 - 29| = 8$$

Midpoint Rectangle Method

For the midpoint method, the function value from the middle of each interval is used.

The middle points are:

$$1.5, 2.5, 3.5$$

The function values are:

$$\begin{aligned} f(1.5) &= 2.25 \\ f(2.5) &= 6.25 \\ f(3.5) &= 12.25 \end{aligned}$$

So, the approximation is:

$$\begin{aligned} I_M &= h[f(1.5) + f(2.5) + f(3.5)] \\ I_M &= 1(2.25 + 6.25 + 12.25) \\ I_M &= 20.75 \end{aligned}$$

The error is:

$$Error = |21 - 20.75| = 0.25$$

Trapezoidal Method

For the trapezoidal method, the area under the curve is approximated by trapezoids instead of rectangles.

The formula is:

$$\begin{aligned}I_T &= h \left[\frac{f(1) + f(4)}{2} + f(2) + f(3) \right] \\I_T &= 1 \left[\frac{1 + 16}{2} + 4 + 9 \right] \\I_T &= 8.5 + 13 \\I_T &= 21.5\end{aligned}$$

The error is:

$$Error = |21 - 21.5| = 0.5$$

Results for Exercise 1

Method	Approximation	Exact Value	Error
Left rectangle	14	21	7
Right rectangle	29	21	8
Midpoint rectangle	20.75	21	0.25
Trapezoidal	21.5	21	0.5

From the table, the midpoint method gave the best result for this example. The left rectangle method underestimated the integral, while the right rectangle method overestimated it. The trapezoidal method was also close to the exact value.

Exercise 2 - Checking the Results with Python

In this part, I used the given Python notebook lab7.ipynb to check the manual calculations from Exercise 1.

The same function was used:

$$f(x) = x^2$$

and the same interval was used:

$$[1, 4]$$

with three intervals.

The numerical results from the code were the same as the manual calculations.

The Python results were:

Method	Python Result
Left rectangle	14
Right rectangle	29
Midpoint rectangle	20.75
Trapezoidal	21.5

This confirms that the manual calculations were correct.

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Lab 7 - Exercise 1 and Exercise 2
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Integral: from 1 to 4 of x^2 dx
Exact analytical value: 21.000000000000004

Left rectangle result: 14.0
Left rectangle error: 7.0000000000000036

Right rectangle result: 29.0
Right rectangle error: 7.9999999999999964

Midpoint rectangle result: 20.75
Midpoint rectangle error: 0.25000000000000355

Trapezoidal result: 21.5
Trapezoidal error: 0.49999999999999645
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Exercise 3 - Numerical Integration of e^{-x^2}

In this exercise, I used the same numerical methods to calculate:

$$\int_{-1}^3 e^{-x^2} dx$$

The function is:

$$f(x) = e^{-x^2}$$

This integral does not have a simple elementary antiderivative, so numerical methods are useful for calculating it.

The reference value used for comparison is approximately:

$$I \approx 1.6330314811$$

I calculated the integral using different numbers of intervals. The aim was to see how the precision changes when the number of intervals increases.

Results for e^{-x^2}

Number of intervals	Left rectangle	Right rectangle	Midpoint rectangle	Trapezoidal
4	1.754075	1.386318	1.664931	1.570197
8	1.709503	1.525625	1.640812	1.617564
16	1.675157	1.583218	1.634956	1.629188
32	1.655057	1.609087	1.633511	1.632072
64	1.644284	1.621299	1.633151	1.632792

Error Table

The exact value used for the error calculation was:

1.6330314811

Number of intervals	Left error	Right error	Midpoint error	Trapezoidal error
4	0.121043	0.246713	0.031900	0.062835
8	0.076471	0.107407	0.007780	0.015468
16	0.042126	0.049813	0.001925	0.003844
32	0.022025	0.023944	0.000480	0.000959
64	0.011253	0.011732	0.000120	0.000240

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Lab 7 - Exercise 3
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Integral: from -1 to 3 of exp(-x^2) dx
Reference value: 1.6330314810719482

Results Table
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Intervals | Left      | Right     | Midpoint  | Trapezoidal
-----
4 | 1.754075 | 1.386318 | 1.664931 | 1.570197
8 | 1.709503 | 1.525625 | 1.640812 | 1.617564
16 | 1.675157 | 1.583218 | 1.634956 | 1.629188
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Error Table
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8 | 0.076471 | 0.107407 | 0.007780 | 0.015468
16 | 0.042126 | 0.049813 | 0.001925 | 0.003844
32 | 0.022025 | 0.023944 | 0.000480 | 0.000959
64 | 0.011253 | 0.011732 | 0.000120 | 0.000240

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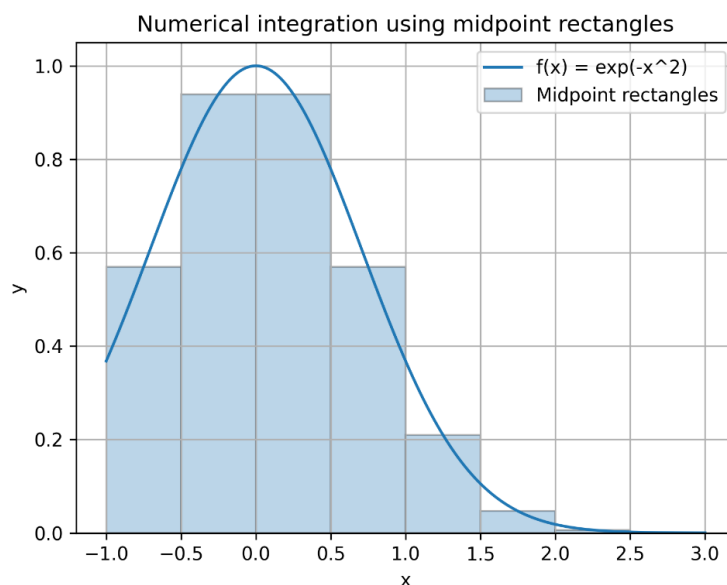
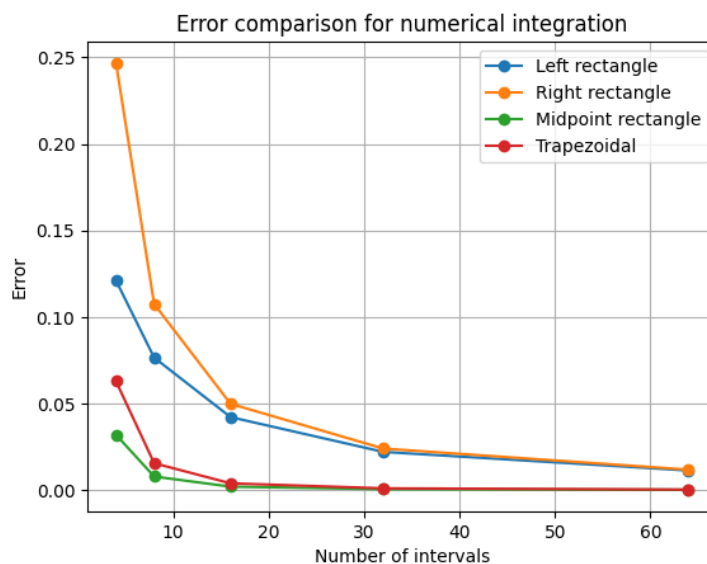
Discussion

From the results, I observed that increasing the number of intervals improves the accuracy of all methods. When the interval width becomes smaller, the numerical approximation becomes closer to the exact value.

The left and right rectangle methods have larger errors compared to the midpoint and trapezoidal methods. The midpoint method gave the smallest error in this example. The trapezoidal method was also quite accurate and became better when the number of intervals increased.

For example, when only 4 intervals were used, the midpoint error was about 0.031900. When 64 intervals were used, the midpoint error decreased to about 0.000120. This shows that the precision strongly depends on the number of intervals.

In general, more intervals give better precision, but they also require more calculations.



Conclusion

In this lab, I learned how to approximate definite integrals by using numerical integration methods. I first calculated the integral of x^2 from 1 to 4 manually using left rectangles, right rectangles, midpoint rectangles, and the trapezoidal method.

The exact value of the integral was:

$$21$$

The best approximation for this case was obtained by the midpoint method:

$$20.75$$

with an error of:

$$0.25$$

Then I checked the calculations using the given Python notebook. The code results matched the manual calculations.

Finally, I calculated the integral of e^{-x^2} from -1 to 3 using different numbers of intervals. I saw that the error decreases when the number of intervals increases. The midpoint and trapezoidal methods were more accurate than the left and right rectangle methods.

Overall, this lab showed that numerical integration can give good approximations, especially when the interval is divided into smaller parts.